

EXERCISE 1 (SQL FUNDAMENTALS)

01. Selecting and Sorting.

1. SELECT *
FROM employees ;

Output :

Id	first-name	last-name	department	salary	hire-date
1	Alice	Green	IT	70000	2020-01-10
2	Brian	Lee	HR	45000	2019-03-22
3	Cathy	Zulu	Finance	65000	2018-07-18
4	David	Mokoena	Marketing	50000	2021-11-05
5	Eva	Maidoo	IT	72000	2017-09-30

City

Johannesburg

Cape Town

Durban

Pretoria

Johannesburg

2. SELECT DISTINCT department
FROM employees ;

Output

department

IT

HR

Finance

Marketing

```

3. SELECT first-name,
           last-name
FROM employees
ORDER BY salary DESC;

```

Output:

first-name	last-name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

```

4. SELECT id,
           first-name,
           last-name,
           salary
FROM employees
ORDER BY salary DESC
LIMIT 3;

```

Output:

id	first-name	last-name	salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

02. Filtering with WHERE.

05. SELECT id,
first-name,
last-name,
department
FROM employees
WHERE department = 'IT' ;

Output ;

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

06. SELECT id,
first-name,
last-name,
department,
salary
FROM employees
WHERE department = 'finance' AND salary > 60000;

Output :

id	first-name	last-name	department	salary
3	Cathy	Zulu	finance	65 000

07. SETE

07. SELECT id,
first-name,
last-name,
department

FROM employees

WHERE department = 'HR' OR department = 'Marketing';

Output:

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

08. SELECT id,
first-name,
last-name,
department

FROM employees

WHERE NOT department = 'IT';

Output:

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

09. SELECT id,
first-name,
last-name,
department

FROM employees

WHERE department IN ('IT', 'HR', 'Finance');

Output:

id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

```

10. SELECT id,
        first-name,
        last-name,
        department,
        salary,
        city
FROM employees
WHERE department = 'IT' AND salary > 65 000
AND city = 'Johannesburg';

```

Output:

id	first-name	last-name	department	Salary	City
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naidoo	IT	72 000	Johannesburg